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REVIEW

# Adhesive Capsulitis: Review of Current Concepts

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## Abstract

**Purpose of Review** This manuscript reviews key concepts surrounding adhesive capsulitis (AC), including its pathophysiology, clinical presentation, diagnosis, and management. This topic is especially important, as AC is a common form of progressive, atraumatic shoulder pain.

**Recent Findings** Recent research has emphasized the growing role of additional imaging modalities for diagnosis, including dynamic ultrasound, as well as advances in conservative management strategies, such as ultrasound deep heat therapy. Additionally, minimally invasive treatment options, including platelet-rich plasma injections, hydrodilatation, and suprascapular nerve blocks, are being actively studied for their potential to provide both short- and long-term pain relief and functional improvements.

**Summary** Accurate diagnosis of adhesive capsulitis requires a thorough history and physical examination to confirm the diagnosis and exclude other etiologies of shoulder pain. Ongoing research aims to refine clinical classification and staging systems to guide more personalized treatment strategies. Advances in imaging and minimally invasive treatment options continue to improve available management options. Future research is needed to refine these treatment approaches and facilitate early and effective treatment for patients with AC.

**Keywords** (max 6): adhesive capsulitis · Frozen shoulder · Shoulder pain · Shoulder capsule · Glenohumeral joint

## Introduction

Adhesive capsulitis (AC), or "frozen shoulder," is a clinical diagnosis best supported by a history of shoulder pain, progressing gradually from acute to chronic, with concomitant decline in joint mobility. Limitations primarily manifest in external rotation, with subsequent stiffness during abduction in the scapular plane, followed by internal rotation and flexion [1]. These restrictions suggest a capsular pattern and are seen in both passive and active range of motion (ROM).

Though the etiology is not well understood, AC has a predilection for middle-aged patients, and more commonly presents in women than in men, which is theorized to be due to hormonal and/or immunological variations [1]. Persons with endocrinological disorders are also more likely to be affected, with nearly a third of those diagnosed being diabetic [2]. Other potential risk factors include hyperlipidemia, cardiovascular disease, thyroid disease [3], and prior trauma and immobility of the shoulder [1]. During the COVID-19 pandemic, one observational study documented a 2.41-fold increase in the incidence of AC and suggested a correlation with psychosocial factors, including anxiety and depression [4, 5].

Whether it is idiopathic or secondary to an underlying condition, AC can cause significant disruption in a patient's functional status. Therefore, early recognition is crucial to improving the likelihood of restoration, which often requires robust, multimodal interventions.

## Pathophysiology

The glenohumeral joint (GHJ) boasts the widest ROM in the human body, as the alignment of the articulating surfaces and its diarthrodial nature facilitate extensive multiaxial movement. The GHJ capsule is a thick, encasing sheath composed primarily of type 1 collagen that attaches to the entire circumference of the anatomical neck of the humerus and to the rim of the glenoid fossa

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[6]. Contraction of this capsule is often implicated in the mechanism of adhesive capsulitis, though the exact disease process remains unclear.

Histologically, the initial phase of AC is marked by upregulation of pro-inflammatory cytokines such as interleukins, tumor necrosis factor, and cyclooxygenase 1 and 2 [7], which stimulate fibroblast proliferation. There is also increased expression of vascular and nerve growth factors, giving rise to subsynovial neoangiogenesis and neoinnervation. This, along with upregulated sensory ion channels, contributes significantly to nociceptive signalling during the early stages of the disease [8].

The evolution of adhesive capsulitis has been linked to persistent, high-volume fibroblast to myofibroblast differentiation, identified by positive staining for α-smooth muscle actin. These myofibroblasts cause remodeling of the extracellular matrix with imbalanced production and packing of immature type III collagen [9], leading to varying degrees of fibrosis within the GHJ capsule, particularly at the rotator interval. Gross findings from arthroscopy in patients with AC demonstrated elements of subacromial fibrosis, proliferative synovitis, and capsular thickening [1], all of which contribute to stiffness of the joint.

## Diagnosis

### History Taking

Comprehensive history taking is imperative in determining the etiology of shoulder pain. The patient should first identify the exact location of pain, which in AC can be generalized to the anterior, lateral, and posterior aspects of the shoulder. The onset of symptoms is commonly progressive over weeks to months and often not sudden. Knowing the past medical history is important in understanding a primary (e.g., idiopathic, type 2 diabetes mellitus, cardiovascular disease, thyroid disorders) versus secondary (e.g., recent trauma, surgery, immobility from a stroke) etiology of AC [10]. In patients with AC, there are underlying chronic inflammatory states or periods of immobility that may be contributing to fibrosis of the joint capsule, and therefore knowing this history is useful in guiding a diagnosis [11]. Another important aspect of the history would be understanding the physical limitations of the patient during this time. Patients will often state that their upper extremity feels "stuck" and that they are unable to perform certain activities of daily living due to limitations in shoulder ROM (i.e. brushing hair, putting on shirt, reaching overhead for an item). Understanding this description is important in order to differentiate other causes of musculoskeletal shoulder pain such as rotator cuff tendinopathy or tear, which may present with pain-limiting ROM without a hard-end "frozen" feel to shoulder movement.

### Physical Examination

Clinical examination is a crucial step in evaluating shoulder pathology and ruling out other focal or referred etiologies of shoulder pain. Physical examination also acts to supplement the medical history in order to guide an accurate diagnosis. Inspection of the shoulder is typically normal unless there is a secondary cause of AC, or the patient is in severe discomfort at rest with the upper extremity in adduction and internal rotation [12]. Both active and passive ROM are painful but also limited due to fibrosis of the GHJ. External rotation of the shoulder is often affected first; however, as the disease progresses, shoulder ROM becomes more globally limited [10]. Therefore, one distinguishing characteristic that supports AC is limited passive shoulder ROM with a firm and painful end-point without obvious deformity [10, 13]. Palpation of the shoulder may present as generalized and nonspecific tenderness to surrounding shoulder musculature and GHJ. Sensation, vascular, and neurologic exams are typically intact, and muscle strength is relatively normal within the presenting degrees of ROM. There are no specific special tests to establish the diagnosis of AC; however, cervical and shoulder special tests may support other etiologies of neck or shoulder pain.

### Classification

Several classification systems for AC have been proposed, but there is no universally accepted standard. The first known published classification system was described by Lundberg, which categorized AC into primary and secondary [14]. Primary AC was defined as total elevation in the shoulder joint of 135 degrees or less, restriction of motion localized to the humero-scapular joint, and no identifiable underlying cause. Secondary AC had similar motion restrictions, but this was due to another diagnosis such as trauma, rheumatoid arthritis, osteoarthritis, or hemiplegia [14]. Nash and Hazleman expanded the secondary category to include other diseases, such as diabetes mellitus and other neurologic conditions [15].

Zuckerman and Rokito surveyed American Shoulder and Elbow Surgeon members to establish consensus on their proposed system [16]. Most respondents agreed with the classification of primary and secondary types, with secondary AC further subdivided into intrinsic, extrinsic, and systemic etiologies [16]. Systemic AC includes diagnoses related to underlying systemic connective tissue disease processes. Extrinsic secondary AC includes diagnoses



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whose pathology is not directly related to the shoulder, while intrinsic describes patients with known pathology of the shoulder joint or surrounding structures. This system allows for improved prognostication based on classification. For example, patients with secondary AC due to insulin-dependent diabetes mellitus may have a more prolonged course of recovery.

Robinson et al. proposed a classification system of primary and secondary etiologies, but separating diabetic AC into a third and independent category [17]. These categories were further defined by specific diagnoses based on the expansion of AC research. Primary AC included idiopathic cases, systemic diseases (e.g., thyroid disorders, hyperlipidemia, COPD), and miscellaneous conditions (e.g., ipsilateral distal limb injury, Dupuytren's disease, ischemic heart disease). Secondary AC included orthopedic and musculoskeletal causes, which was further categorized into traumatic (e.g., soft tissue injury, dislocation, fracture) and non-traumatic (e.g., rotator cuff tear, tendinopathy, early osteoarthritis). Diabetic AC was considered a distinct category due to its unique course and management challenges [17].

While these classification systems have focused on diagnosis and treatment, others have also been proposed to guide rehabilitation. Kelley et al. proposed a system based on irritability level, where those with low irritability have less pain, little to no pain at end ROM, and low disability, while those with high irritability have significant pain, limited passive ROM with pain and guarding, and greater disability [18]. Patients with low irritability generally report stiffness as their chief complaint, while patients with high irritability report pain. Due to the natural history of AC, patients in early stages generally have high irritability, while those in later stages have low irritability [18]. Kelley et al. also proposed an impairment-based model, which includes medical screening, differential diagnosis of musculoskeletal impairments, tissue irritability level, and targeted intervention strategies for pain and mobility deficits [19].

## Clinical Staging

There is no standardized clinical staging system. However, the four-stage model by Hannafin and Chiaia is widely used [20]. Based on clinical, arthroscopic, and histologic findings, the stages span approximately 24 months. Stage 1 (prefreezing) lasts for 3 months, with pain at rest and end of shoulder ROM due to hypervascular synovitis. Stage 2 (freezing) lasts from 3 to 9 months, characterized by persistent pain and progressive loss of motion due to scar formation and capsular fibroplasia. Stage 3 (frozen) is from 9 to 15 months, where patients are relatively pain-free but have significant loss of motion due to dense scar formation of the capsule. Stage 4 (thawing) lasts from 15 to 24 months, where patients have minimal pain and progressive improvement in ROM [20, 21]. There are also some proposed clinical staging guidelines based on degree of external rotation (ER) ROM, including mild for ER greater than or equal to 45 degrees, moderate for ER between 11 and 45 degrees, and severe for ER less than or equal to 10 degrees [17].

## Imaging Studies

Magnetic resonance imaging (MRI) and magnetic resonance arthrography (MRA) are the routine imaging modalities for diagnosis of AC. While contrast-enhanced studies have a higher sensitivity and specificity for imaging findings consistent with AC, these are less frequently performed due to their more invasive nature [22]. One of the earliest studies describing MRI findings in AC by Emig et al. showed that shoulder capsule and synovium thickness measurements of greater than 4 mm can be diagnostic of AC with 70% sensitivity and 95% specificity [23]. This criteria is still widely used today [2]. This is best viewed on MRI coronal oblique T2-weighted images, as T1-weighted images may overestimate capsular thickening given fat suppression [22]. Other key findings on MRI/MRA include thickening of the rotator interval capsule and coracohumeral ligament (CHL), which is best visualized on sagittal oblique images. These findings are highly specific (95%) but have low sensitivity (64%), with MRA generally more sensitive than MRI [22]. A CHL thickness of greater than 3 mm provides the highest accuracy for diagnosis by MRA. Rotator interval synovitis may also be demonstrated by intermediate-to-low T1 thickening, hyperintense edema, and post-contrast enhancement of the rotator interval capsule. Signal hyperintensity of the axillary pouch and inferior glenohumeral ligamentous complex on T2-weighted fat-suppressed sequences also showed high sensitivity (82–94%) and specificity (88%) [22]. Complete obliteration of the subcoracoid fat triangle, or the triangular space bounded by the coracoid process, joint capsule, and CHL, is poorly sensitive (32%) but highly specific (100%) for diagnosis of AC [22].

Some studies have correlated MRI/MRA findings with AC stages as described by Hannafin and Chiaia [20]. T2 hyperintensity of the joint capsule and axillary capsule thickness is more frequently seen in stages 1 and 2 compared to later stages, reflecting inflammation and edema of the capsule as well as early fibrosis. Other findings, including effusion around the long head of biceps, subcoracoid fat obliteration, and CHL thickness, were not stage-specific [21].

CT arthrography demonstrates similar findings to MRI/MRA, including axillary pouch thickening, CHL thickening, rotator interval obliteration, and decreased axillary recess width [2, 22]. These studies are most



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reliable when limited GHJ capacity is confirmed during contrast injection [2]. Fluoroscopic-guided glenohumeral arthrography was previously used for diagnosis, but this has been largely replaced by MRI/MRA. Classic findings include decreased capsular distension (less than 10 ml, compared to 14 ml or greater in a normal joint), obliteration of and filling defects in the axillary recess, and early extracapsular extension (extravasation) of contrast through areas of weakened joint capsule including to the biceps tendon sheath [22]. Plain radiographs are typically used to exclude other causes of shoulder pain (e.g., glenohumeral arthritis, calcific tendinosis) rather than to diagnose AC. Occasionally, osteopenia may be seen due to disuse and inflammation [22].

Ultrasound is becoming an increasingly utilized modality due to its accessibility, cost-effectiveness, and ability to assess patients with contraindications to MRI/MRA. Findings are similar to those noted on MRI/MRA, including increased thickness of the CHL (greater than or equal to 3 mm), inferior glenohumeral capsule (4 mm), and axillary recess capsule (4.4 mm) [22]. Doppler signal and hypoechoic soft tissue in the rotator cuff interval may also support diagnosis [2]. Another key advantage of ultrasound is the ability to do dynamic studies. Restricted subacromial gliding of the supraspinatus tendon is highly predictive of decreased intra-articular injection volume [22]. A relatively new technique is performing ultrasound arthrography using contrast-enhanced ultrasound (CEUS) and microbubble contrast agents, which allows for assessment of capsular changes. Filling defects likely related to irregular thickening and distortion of capsular tissue and synovium showed 91% sensitivity and 87% specificity [22]. Hyperechoic microbubbles on debris in the joint or on the capsule's subsynovial layer were characteristic of AC with 76% sensitivity and 78% specificity. Both of these findings combined were found to be more sensitive than standard ultrasound for detecting AC. Despite promising results, the role of ultrasound in the diagnosis of AC is still not well-defined. Therefore, it is not yet recommended by the European Society of Musculoskeletal Radiology as first-line for evaluation of patients with suspected AC [2, 22].

## Management

Management of AC should be tailored to each patient, and will usually depend on symptom severity, disease stage, comorbidities, and their treatment preferences. This is generally a comprehensive approach with stepwise treatments that can start from simple and conservative to more invasive interventions.

### Oral Pharmacological Treatments

Initial treatment often begins with pharmacologic interventions aimed at reducing inflammation and controlling pain, particularly in the early phases. Nonsteroidal anti-inflammatory drugs (NSAIDs) are the most commonly used; however, when used in isolation, their efficacy is limited. A 2024 systematic review and meta analysis evaluated two randomized controlled trials (RCTs) for efficacy of oral NSAIDs and found that it only provided mild relief, with limited impact on ROM or long-term outcomes compared to placebo [24]. However, it is still a good initial choice to control pain and facilitate tolerance of physical therapy (PT).

### Therapy

Rehabilitation is the most consistent treatment over the years and is central to managing AC. Therapy should begin as early as possible. It is often used alone or in combination with other treatments. A 2021 systematic review analyzed physiotherapy interventions including mobilization, stretching exercises, proprioceptive neuromuscular facilitation (PNF), and strengthening for AC and found significant improvements in pain, ROM, and functional status when compared to placebo or no treatment [25].

A 2024 systematic review evaluated muscle energy techniques (MET) and joint mobilization techniques (JMT) for improving ROM, pain, and function in patients with AC. It included a total of 9 RCTs and 439 participants with different therapy interventions such as Mulligan's mobilization with movement, Spencer's MET, Maitland mobilization, and mobilization using mechanical apparatuses. The majority of the studies showed good outcomes; however, the evidence was low to very low due to heterogeneity and methodological limitations. Joint mobilization techniques and MET both showed greater improvements in ROM, pain, and function when compared to stretching or conventional PT alone. Mulligan's and Spencer's techniques were most effective with variations in different studies, and in general, some suggested better pain relief with MET and superior ROM outcomes with JMT. Despite limited quality evidence, both MET and JMT are promising physiotherapy approaches for AC management [26].

A 2025 RCT evaluated Gong's Mobilization compared to Mulligan's Mobilization with Movement (MWM) in patients with AC. Gong's Mobilization focuses on correcting humeral head position during dynamic shoulder motion addressing muscular and capsular tightness, while MWM focuses on pain free joint mobilization of the joint with therapist assisted joint glide. Both techniques helped reduce pain and improve ROM, but patients in the Gong's Mobilization group experienced greater gains in shoulder function,



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particularly in abduction and external rotation. While pain relief was similar between the groups, those treated with Gong's approach had better overall improvements in disability scores, which suggests that Gong's Mobilization may be more effective in restoring shoulder function AC patients [27].

## Modalities

Ultrasound deep heat therapy (UST) has been used as an adjunct treatment for AC. A 2022 meta-analysis found that UST significantly improved pain outcomes when compared to placebo when it was used in addition to PT or exercise; however, it did not consistently improve ROM or disability scores. When compared with other interventions, corticosteroid injections led to improved ROM and comparable pain scores but with faster onset of relief. Platelet-rich plasma (PRP) injections showed better outcomes in both pain and function. Furthermore, UST was comparable to cryotherapy in terms of pain relief, and was less effective than shockwave therapy, especially for improving shoulder function. These findings suggest that UST may be useful mainly for pain relief when used in combination with rehabilitation, but its benefits for mobility and long-term function remain limited [28].

A 2023 RCT included 40 patients with AC and compared the effects of high-intensity laser therapy (HILT) versus low-level laser therapy (LLLT) combined with stretching exercises for both treatment arms. Both groups showed significant improvements in pain and functional scores, with the HILT group having greater improvements after three weeks. Neither group showed meaningful gains in shoulder ROM [29].

## Injectates

Intra-articular (IA) corticosteroid injections are the most commonly used injectate for AC. A systematic review of early-stage treatment found that corticosteroids show sustained improvement compared to placebo, with better short-term pain relief and functional gains when compared to PT alone. However, there is generally no difference in long-term outcomes [1]. Similarly, corticosteroids showed superior short-term results in terms of pain and function when compared to hyaluronic acid (HA), with both treatments showing similar outcomes by 6 to 12 weeks. This suggests that HA may be a reasonable option for patients who cannot receive steroids secondary to comorbidities or if multiple injections are required [30].

Platelet-rich plasma (PRP) has become increasingly popular for its potential anti-inflammatory and regenerative properties for managing AC. A meta-analysis of 14 RCTs, including over 1,000 patients, found that PRP led to significant improvements in shoulder pain and disability index (SPADI), disability of arm, shoulder, and hand (DASH), and VAS scores, as well as in both active and passive ROM when compared to placebo and other injections including steroid and HA. When compared to hydrodilatation, both treatments led to improvements in pain, function, and ROM. However, PRP demonstrated superior outcomes in several secondary measures, including active shoulder flexion and abduction at 24 weeks and SPADI scores across early and mid-term follow-up. It is worth noting that the therapeutic advantage of PRP is most prominent in the first 12 to 24 weeks, with comparable long-term outcomes between the two treatments [31]. Several other studies comparing PRP to corticosteroids showed more favorable outcomes with PRP. For example, Barman et al. reported greater pain relief and ROM improvements at 12 weeks, while Shahzad et al. similarly found better ROM gains with PRP. Based on these studies, PRP is generally well-tolerated with fewer adverse effects than corticosteroids, and consistently provides favorable outcomes [31].

## Hydrodilatation

Hydrodilatation involves distension of the GHJ capsule with the goal of mechanically disrupting the capsular adhesions to decrease pain and improve ROM. It commonly uses a combination of saline, corticosteroid, and anesthetic under ultrasound or fluoroscopic guidance. A systematic review and meta-analysis of 8 RCTs compared hydrodilatation (HD) with conventional treatment versus conventional treatment alone in patients with AC. Patients receiving HD in combination with standard care had significantly greater reductions in pain and improved shoulder function compared to those receiving standard care alone. However, this review found high heterogeneity across studies secondary to differences in intervention protocols and disease severity [32].

Different techniques for performing hydrodilatation have been explored, particularly anterior versus posterior approaches. The posterior approach is commonly used due to easier accessibility and typically involves an in-plane ultrasound-guided injection into the GHJ. In contrast, the anterior approach targets the rotator interval between the biceps tendon and coracohumeral ligament. An RCT from 2020 compared these two methods and found that the anterior approach led to greater improvements in external rotation and pain relief. This difference is thought to be due to a larger concentration of steroid reaching the rotator interval and tissues such as the anterior capsule and coracohumeral ligament with the anterior approach and greater loosening of adhesions from the pressure of injection into this more compact space [33]. Similar reasoning was attributed to the differences found between approaches in a meta-analysis



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evaluating steroid injections and one which evaluated RCTs of steroid injections, with hydrodilatation included in some of the studies [34, 35]. Another study evaluated capsule-preserving versus capsule-rupturing hydrodilatation. In this study of 149 patients, both techniques led to improvements in pain and function. However, patients in the capsule-preserving group had consistently better DASH and VAS scores at 1, 3, and 6 months. Capsular rupture was also associated with worse outcomes, potentially due to steroid leakage into nearby tissues [36].

### Suprascapular Nerve Block +/- Hydrodilatation

Suprascapular nerve block (SSNB) is also an appealing minimally invasive option. This is typically performed under ultrasound guidance, targeting the primary sensory innervation of the shoulder, and can significantly reduce pain. A recent meta-analysis of 12 RCTs found SSNB outperformed IA steroid injections and PT in terms of pain relief and functional improvement, particularly for external rotation [37]. Combining HD with SSNB has also been investigated. A recent retrospective study with 150 patients with AC underwent HD, SSNB, and immediate post-injection PT. The HD technique used a posterior in-plane medial-to-lateral approach, delivering a mixture of saline, corticosteroid, and anesthetic directly into the GHJ under ultrasound guidance. Results showed statistically significant improvements in active forward flexion, external rotation, and VAS scores at both short-term and follow-up intervals [38]. Interestingly, in one prospective RCT with 48 patients comparing SSNB alone versus SSNB with HD, there were no statistically significant differences found between groups in most outcome measures over a 3-month follow-up period. Therefore, SSNB alone may be a minimally invasive option for treating AC [39].

### Surgical Treatments

For patients with persistent symptoms despite conservative and interventional treatments, surgical options may be considered. Arthroscopic capsular release (ACR) and manipulation under anesthesia (MUA) are the most common surgical strategies. A recent meta-analysis comparing these two approaches found no significant differences in VAS scores, external rotation, or overall complication rates at 3, 6, and 12 months. However, ACR demonstrated slightly better forward flexion outcomes at 3 and 6 months [40]. This may be due to the controlled nature of capsular release under direct visualization. Nonetheless, both procedures appear to provide comparable long-term results, and the choice may depend on patient comorbidities, surgeon expertise, and institutional resources.

### Other Emerging Techniques

Emerging techniques specifically targeting the CHL are being used for the treatment of AC, such as steroid injections into the ligament [41], percutaneous needling, and CHL release with tenotomy [42]. Further research is required to understand the efficacy and long-term outcomes of these interventions.

## Conclusion

Adhesive capsulitis is a common cause of shoulder pain functional limitation. Recognizing the risk factors that predispose individuals to AC, understanding its pathophysiology and natural history, and accurately identifying affected patients is integral to early and effective management. While there are no standardized classification or clinical staging systems, research in this area continues to evolve, aiming to better define its stages in order to tailor treatment accordingly. Advances in imaging, particularly the growing use of ultrasound, have enhanced diagnostic capabilities. Additionally, research continues to grow on minimally invasive treatment options, including PRP, HD, and SSNB. Further research is needed to refine these approaches and improve functional outcomes for patients with AC.

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**Competing Interests** The authors declare no competing interests.

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